

Subsequences

Definition

- Let $\{u_n\}$ be a real sequence.
- $\{r_n\}$ be a strictly increasing sequence of natural numbers, i.e.,
 $r_1 < r_2 < r_3 < \cdots < r_n < \cdots$.
- Then the sequence $\{u_{r_n}\}$ is said to be a **subsequence** of the sequence $\{u_n\}$.
- The elements of the subsequence $\{u_{r_n}\}$ are

$$u_{r_1}, u_{r_2}, u_{r_3}, \dots, u_{r_n}, \dots$$

Definition

- We start with an original sequence.
- We choose certain positions (indices).
- We keep the elements in the same order.
- The result is a subsequence.

Definition

Original sequence: $a_1, a_2, a_3, a_4, a_5, a_6, \dots$

Subsequence: $a_{n_1}, a_{n_2}, a_{n_3}, a_{n_4}, \dots$

where $1 \leq n_1 < n_2 < n_3 < n_4 < \dots$

Visual Example: Original Sequence

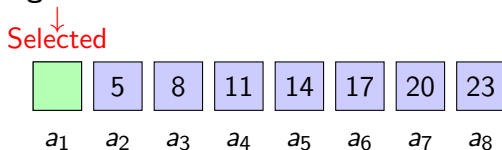
Original Sequence:

2	5	8	11	14	17	20	23
a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8

- This is our starting sequence
- We'll create a subsequence by selecting certain terms
- The order of selected terms must be preserved

Creating a Subsequence - Step 1

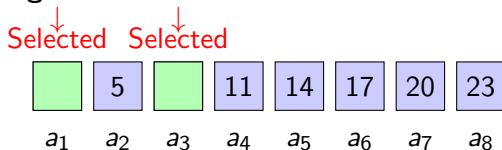
Selecting Terms:



- First selection: $a_1 = 2$
- We choose the first term
- Index: $n_1 = 1$

Creating a Subsequence - Step 2

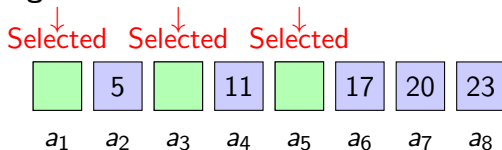
Selecting Terms:



- Second selection: $a_3 = 8$
- We skip a_2 and choose a_3
- Index: $n_2 = 3$

Creating a Subsequence - Step 3

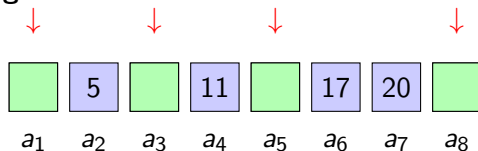
Selecting Terms:



- Third selection: $a_5 = 14$
- We skip a_4 and choose a_5
- Index: $n_3 = 5$

Creating a Subsequence - Step 4

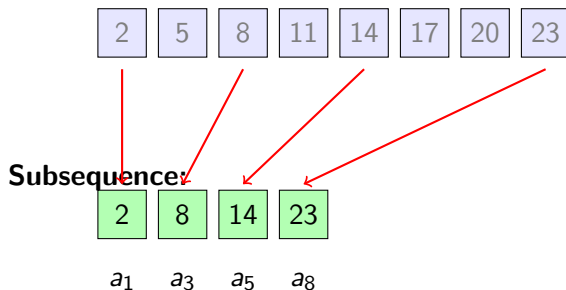
Selecting Terms:



- Fourth selection: $a_8 = 23$
- We skip a_6 , a_7 and choose a_8
- Index: $n_4 = 8$

The Resulting Subsequence

Original Sequence:



Subsequence: a_1, a_3, a_5, a_8
 $= 2, 8, 14, 23$

Another Example: Even-indexed Terms

Original Sequence: $a_n = n^2$

1		9		25		49	
a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8

Subsequence: Even indices only

4	16	36	64
a_2	a_4	a_6	a_8

Original: 1, 4, 9, 16, 25, 36, 49, 64, ...

Subsequence: $a_2, a_4, a_6, a_8, \dots = 4, 16, 36, 64, \dots$

Why Study Subsequences?

Importance in Analysis

- **Convergence:** If a sequence converges, all its subsequences converge to the same limit
- **Limit Points:** Subsequences help to identify limit points.
- **Alternative Approach:** Sometimes it is easier to study a sequence through its subsequences

Examples

1. Let $\{u_n\} = \{\frac{1}{n}\}$ and $r_n = 2n$ for all $n \in \mathbb{N}$ i.e., $\{r_n\} = \{2, 4, 6, \dots\}$. Then

$$\{u_{r_n}\} = \{u_2, u_4, u_6, \dots\} = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots\right\}.$$

So $\{u_{r_n}\}$ is a subsequence of $\{\frac{1}{n}\}$.

Examples

2. Let $u_n = \frac{1}{n}$ and $r_n = 2n - 1$ for all $n \in \mathbb{N}$. Then

$$\{u_{r_n}\} = \{u_1, u_3, u_5, \dots\} = \left\{1, \frac{1}{3}, \frac{1}{5}, \dots\right\}.$$

So $\{u_{r_n}\}$ is a subsequence of $\left\{\frac{1}{n}\right\}$.

3. Let $\{u_n\} = \{(-1)^n\}$ and $\{r_n\} = \{2n\}$ for all $n \in \mathbb{N}$.

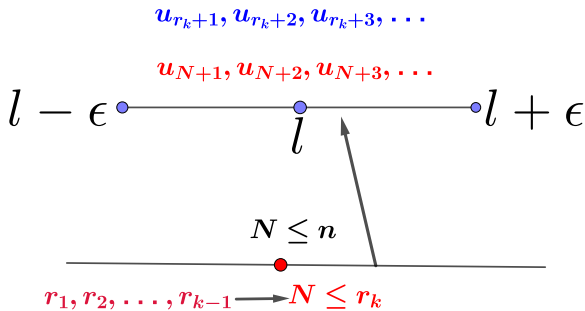
Then

$$\{u_{r_n}\} = \{u_2, u_4, u_6, \dots\} = \{1, 1, 1, \dots\}.$$

So $\{u_{r_n}\}$ is a subsequence of $\{(-1)^n\}$.

Theorem

If a sequence $\{u_n\}$ converges to l , then every subsequence of $\{u_n\}$ also converges to l .



Proof of Theorem

Suppose $\lim u_n = l$.

Then for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|u_n - l| < \epsilon \quad \text{for all } n \geq N.$$

Now let $\{u_{r_n}\}$ be any subsequence of $\{u_n\}$.

Since $\{r_n\}$ is strictly increasing, then $\exists$ a natural number k such that $r_m \geq N$ for all $m \geq k$. Hence,

$$|u_{r_n} - l| < \epsilon \quad \forall n \geq k$$

Therefore, $\lim u_{r_n} = l$.

Thus every subsequence of $\{u_n\}$ converges to the same limit l .

Note

- If there exist two different subsequences $\{u_{r_n}\}$ and $\{u_{k_n}\}$ of a sequence $\{u_n\}$ such that $\{u_{r_n}\}$ and $\{u_{k_n}\}$ converge to two different limits, then the sequence $\{u_n\}$ is not convergent.

Consider the sequence:

$$u_n = (-1)^n$$

Consider two subsequences of $\{u_n\}$:

- For even indices: let $r_n = 2n$, so $u_{r_n} = (-1)^{2n} = 1$. This subsequence converges to 1.
- For odd indices: let $k_n = 2n - 1$, so $u_{k_n} = (-1)^{2n-1} = -1$. This subsequence converges to -1 .

The original sequence $\{u_n\}$ is not convergent.

If a sequence $\{u_n\}$ has a divergent subsequence, then $\{u_n\}$ is divergent.

Theorem

If the subsequences $\{u_{2n}\}$ and $\{u_{2n-1}\}$ of a sequence $\{u_n\}$ converge to the same limit l , then the sequence $\{u_n\}$ is convergent and $\lim u_n = l$.

Proof of Theorem

Let $\epsilon > 0$. Since $\lim u_{2n} = l$, there exists a natural number k_1 such that

$$|u_{2n} - l| < \epsilon \quad \text{for all } n \geq k_1.$$

Since $\lim u_{2n-1} = l$, there exists a natural number k_2 such that

$$|u_{2n-1} - l| < \epsilon \quad \text{for all } n \geq k_2.$$

Let $k = \max\{k_1, k_2\}$. Then for all $n \geq 2k - 1$,

$$l - \epsilon < u_{2n} < l + \epsilon, \quad l - \epsilon < u_{2n-1} < l + \epsilon.$$

Thus $l - \epsilon < u_n < l + \epsilon$ for all $n \geq 2k - 1$. Therefore $\lim u_n = l$.

Converse

If two subsequences of a sequence converge to the same limit l , the original sequence may not be convergent."

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If two subsequences of a sequence converge to the same limit l , the original sequence may not be convergent."

Consider the sequence $\{u_n\}$ defined by:

$$u_n = \begin{cases} 1 & \text{if } n \text{ is even} \\ n & \text{if } n \text{ is odd} \end{cases}$$

Define two subsequences:

- 1 Let $r_n = 2n$. Then $u_{r_n} = u_{2n} = 1$ for all n . This subsequence converges to 1.
- 2 Let $k_n = 4n$. Then $u_{k_n} = u_{4n} = 1$ for all n . This subsequence also converges to 1.

However, the entire sequence $\{u_n\}$ does not converge. For odd n , $u_n = n \rightarrow \infty$, and for even n , $u_n = 1$. Thus, the sequence is unbounded and divergent. Therefore, even though two subsequences converge to the same limit 1, the original sequence is not convergent.